

Lab 8: Aliasing and Harmonic Analysis

- Review
 - Discrete sampling
 - * Problem (Aliasing)
 - A 10 Hz sine wave sampled at 8 Hz appears to be a 2 Hz sine wave.
 - A 10 Hz sine wave sampled at 9 Hz appears to be a 1 Hz sine wave.
 - A 10 Hz sine wave sampled at 10 Hz appears to be constant
 - A 10 Hz sine wave sampled at 11 Hz appears to be a 1 Hz sine wave. (see, for example, [1])
 - A 10 Hz sine wave sampled at 12 Hz appears to be a 2 Hz sine wave.
 - * Sampling rate theorem
 - “This theorem simply states that the sampling rate must be greater than twice the highest-frequency component of the original signal in order to reconstruct the original waveform correctly [1].”
 - * Fourier Series
 - A periodic signal $f(t)$ with period T (or an arbitrary signal considered from time $t = 0$ to time $t = T$) can be represented as an infinite sum of sine and cosine functions.

$$f(t) = a_0 + \sum_{n=1}^{\infty} \{a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)\} \quad (1)$$

where

$$\omega_0 = \frac{2\pi}{T} \quad (2)$$

$$a_0 = \frac{1}{T} \int_0^T f(t) dt \quad (3)$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt \quad (4)$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t) dt \quad (5)$$

- The amplitudes a_n and b_n represent the frequency contribution to the signal at angular frequency $n\omega_0$
 - Example: square wave
 - * Fourier Transform

- We will define the Fourier Transform mathematically as

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt \quad (6)$$

(Note, there are several definitions for Fourier Transform that differ by a factor of $\frac{1}{2\pi}$ or $\frac{1}{\sqrt{2\pi}}$, for example.)

- The Fourier Transform $F(\omega)$ describes the frequency content of a signal and can be thought of as a generalization of the Fourier Series for infinite domains
- For many of the functions we deal with, the Fourier transform is related to the Laplace transform. Simply replace s with $j\omega$.
- * Discrete Fourier Transform (DFT)
 - The Discrete Fourier transform can be used to approximate the Fourier transform of a signal when it is sampled N times at intervals of Δt
 - Mathematical definition

$$F(k\Delta f) = \sum_{n=0}^{N-1} f(n\Delta t)e^{-j(2\pi k\Delta f)n\Delta t}, \quad k = 0, 1, 2, \dots, N-1 \quad (7)$$

- T is the period of time over which samples are taken. $\Delta f = \frac{1}{T}$, $\Delta t = \frac{T}{N}$. The DFT is most easily interpreted when the signal is periodic and there are an integer number of periods within T .
- FFT - is a fast computer algorithm used to calculate the DFT
- * Folding diagram (page 106, Wheeler and Ganji [1])
 - A 4 Hz sine wave sampled at 10 Hz appears to be a 4 Hz sine wave.
 - A 5 Hz sine wave sampled at 10 Hz appears to be a 5 Hz sine wave.
 - A 6 Hz sine wave sampled at 10 Hz appears to be a 4 Hz sine wave.
 - A 7 Hz sine wave sampled at 10 Hz appears to be a 3 Hz sine wave.
 - A 9 Hz sine wave sampled at 10 Hz appears to be a 1 Hz sine wave.
 - A 10 Hz sine wave sampled at 10 Hz appears to be constant.
 - A 11 Hz sine wave sampled at 10 Hz appears to be a 1 Hz sine wave.

- * Leakage
 - Often, one doesn't know the frequency of the signal to be measured and there isn't an integer number of periods in the DFT domain. In such a case "... significant coefficients at frequencies not present in the original signal [1]" will appear in the DFT.
 - Windowing function - attenuates the signal at the beginning and end of the sample window. Used to reduce leakage. There are many common windowing functions, including: Hann (Hanning) and Hamming

- Example: 25 samples of a 10 Hz sine wave sampled at 100 Hz.
- * White noise
 - Fourier transform of white noise is constant - contains equal amplitude at all frequencies
 - Magnitude of Fourier transform of piecewise constant noise is constant up to about $w_c = \frac{2}{dt}$.
 - Magnitude of discrete Fourier transform of piecewise constant noise is constant.
 - Fourier transform of a system with transfer function $T(s)$ and white noise as the input will yield the frequency response plot shifted vertically by an amount dependent upon the amplitude of the noise.
- Equipment
 - National Instruments USB 6341(link)
 - Breadboard with triple power supply
 - Sweep/function generator (link)
 - Additional (old) function generator
 - Oscilloscope (link)
 - 741 op amp (link)
 - * $V_S^+ = 6\text{ V}$
 - * $V_S^- = -6\text{ V}$
 - Resistors: 1k Ω , 100k Ω
 - LED
 - Capacitors: 0.1 μF
- Imagine that a project you are working on requires you to measure the voltage output of an accelerometer attached to a car part. A simple model suggests that the vibrations you are interested in will be periodic with a frequency of somewhere between 50-150 Hz. You expect the measurements to be noisy, but vibrations with frequencies over about 300 Hz are not relevant to the project you are working on. Also, you need to collect at least 1 second of continuous data. **The objective of this lab is to develop guidelines to assist you in: (i) setting up the data acquisition system (e.g., selecting a daq card, sample rate, and filter) and (ii) interpreting signals obtained from the data acquisition system.**
 - Measure the frequency content of sine and square waves
 - * Use LabView, Matlab, or Python to plot the FFT of your signal

- LabView:
Express>>Signal Analysis>>Spectral
- Matlab:
X=fft(x);
f=(0:N-1)/T;
- Python:
f=np.fft.fftfreq(N,dt)
X=np.fft.fft(x)
- * Set up
 - 500mV pk-pk wave
 - Number of samples: 1000
 - Sample rate: 1000
- * Save FFT data for the following cases
 - 100 Hz sine wave
 - 400 Hz sine wave
 - 600 Hz sine wave
 - 900 Hz sine wave
 - 100 Hz square wave
 - 400 Hz square wave
 - 600 Hz square wave
 - 900 Hz square wave
- * Explain:
 - the location and magnitude of peaks in your FFT data
 - the difference between the FFTs for the the sine and square waves
 - the changes in FFT with changes in frequency of the signal
- Measure the frequency content with increased sample rate
 - * Set up
 - 500mV pk-pk wave
 - Number of samples: 10,000
 - Sample rate: 10,000
 - * Save FFT data for the following cases
 - 100 Hz sine wave
 - 400 Hz sine wave
 - 600 Hz sine wave
 - 900 Hz sine wave
 - 100 Hz square wave
 - 400 Hz square wave
 - 600 Hz square wave
 - 900 Hz square wave
 - * Explain:
 - the difference between your FFT data for 1000 samples per second and 10,000 samples per second

- Acquire a non integer number of periods of the signal
 - * Set up
 - 100 Hz 500mV pk-pk sine/square wave
 - Number of samples: 1005
 - Sample rate: 1000
 - * Save FFT data for each of the following windowing functions (ideally, use the exact same data rather than collecting new data for each window):
 - Rectangular
 - Hann (or Hanning)
 - Hamming
 - Other
 - * Explain:
 - the location and magnitude of peaks in your FFT data
 - the difference between the FFTs for the the sine and square waves
 - the changes in FFT with changes in frequency of the signal
 - * Compare the results for different windowing functions
- Measure the frequency content of superimposed signals with and without a first order (passive) filter
 - * Modify your code to acquire two channels simultaneously
 - * Build an op amp summer to add two inputs
 - * Filter the output of the summer with a passive low pass filter with $R=1k\Omega$ and $C=1.0\mu F$. You might want to measure (and record) the resistors and capacitors.
 - * Set up
 - input 1: 500mV pk-pk 100 Hz sine wave
 - input 2: 500 mV pk-pk sine wave (frequencies given below)
 - Number of samples: 1000
 - Sample rate: 1000
 - * Save FFT data with channel 1 connected to the output of the summer and channel 2 connected to the output of the filter for the following cases
 - input 2: 100 Hz
 - input 2: 400 Hz
 - input 2: 600 Hz
 - input 2: 900 Hz
 - input 2: 925 Hz
 - * Explain:
 - the difference between FFT data for the two channels. What is the effect of the filter? Why might you choose to use a filter?

- the interesting response (in the time domain) in channel 1 when input 2 is set to 100 Hz.
- Measure the frequency content of superimposed signals with and without a second order (active) filter
 - * Filter the output of the summer with an active low pass filter with $R_1 = R_2 = R_B = 1k\Omega$ and $C_1 = C_2 = 1.0\mu F$, $R_A = 585\Omega$ (Op amp circuit 1 or 2 from lab 4). You might want to measure (and record) the resistors and capacitors.
 - * Set up
 - input 1: 500mV pk-pk 100 Hz sine wave
 - input 2: 500 mV pk-pk sine wave (frequencies given below)
 - Number of samples: 1000
 - Sample rate: 1000
 - * Save FFT data with channel 1 connected to the output of the summer and channel 2 connected to the output of the filter for the following cases
 - input 2: 100 Hz
 - input 2: 400 Hz
 - input 2: 600 Hz
 - input 2: 900 Hz
 - input 2: 925 Hz
 - * Explain the FFT data for both channels 1 and 2. What is the effect of the filter? Why might you choose to use a filter?
 - * Compare the output for the first order passive filter (above) to that of this second order active filter?
- Obtain the frequency response plot of a second order system using the DAQ board to generate and approximation to white noise as the input
 - * system: Op amp circuit 1 or 2 from lab 4. You might want to measure (and record) the resistors and capacitors.
 - * Set up
 - Analog Output (input to the system): random noise
 - Analog Input (output of the system):
 - Number of samples: large
 - Sample rate: large
 - * Plot the magnitude of the FFT as a function of frequency (average over many runs)
- Lab report:
 - Your lab report should follow the required lab report structure for this class.

- In the body of your report, you should build a strong case for your objectives. Imagine that a project you are working on requires you to measure the voltage output of an accelerometer attached to a car part. A simple model suggests that the vibrations you are interested in will be periodic with a frequency of somewhere between 50-150 Hz. You expect the measurements to be noisy, but vibrations with frequencies over about 300 Hz are not relevant to the project you are working on. Also, you need to collect at least 1 second of continuous data. **The objective of this lab is to develop guidelines to assist you in:**

- * **(i) setting up the data acquisition system**

- What sample rate would you recommend? Why?
- Would you recommend using a filter? Why?
- Justify all of your recommendations using the measurements you obtained in lab.

- * **(ii) interpreting signals obtained from the data acquisition system.**

- How do you recommend analyzing/processing/interpreting the data you collect with your proposed setup? Why?
- How will the sample rate change your interpretation/analysis?
- If you recommend a specific filter (or no filter at all), how will its presence (or absence) change your interpretation/analysis?
- Do you recommend using a windowing function? Why?
- Justify all of your recommendations using the measurements you obtained in lab.

- Include the following in the Appendix:

- * This lab outline (required)
- * Data you collected during lab (required)
- * Screenshot of each LabView block diagram and/or Matlab scripts used to acquire/process data (required)
- * Responses to the questions throughout the lab outline. (optional)
- * Any other details (data, derivations, calculations, experimental procedure, screen captures etc.)

(The purpose of the Appendix in these reports is to: (i) serve as somewhat of a lab notebook, (ii) demonstrate to the grader what you did during lab, and (iii) hold material that didn't fit into the body of the report but would be necessary to repeat the experiments or justify the claims in the report.)

References

- [1] Anthony J. Wheeler and Ahmed R. Ganji. *Introduction to Engineering Experimentation, 3rd Ed.* Prentice Hall, 2010.